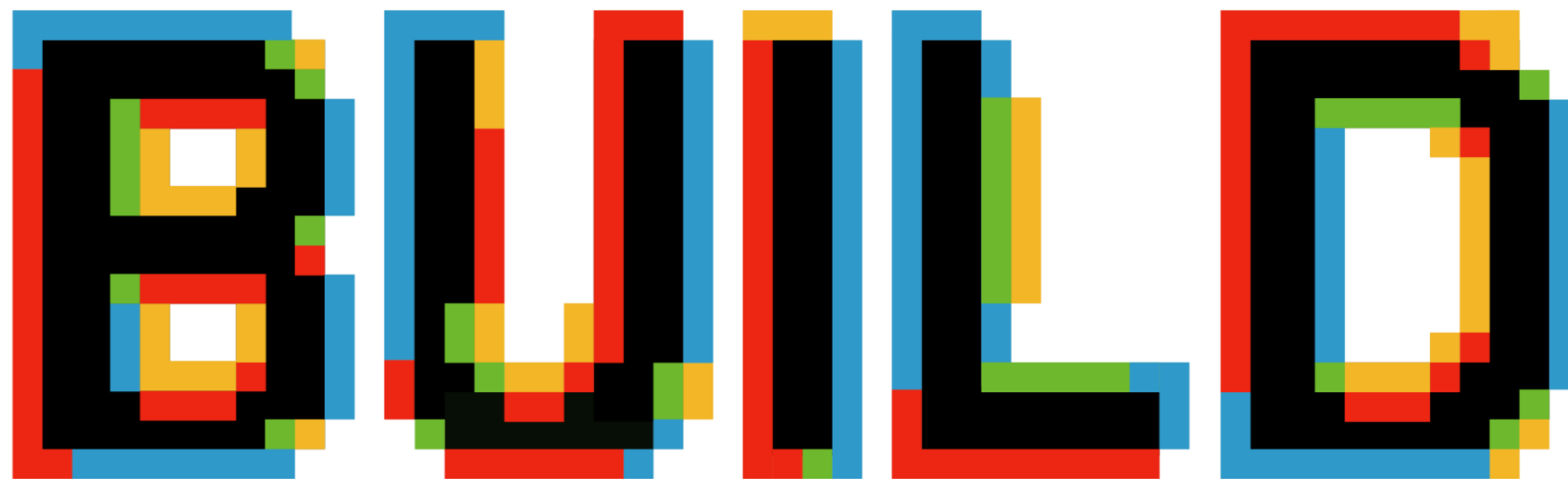
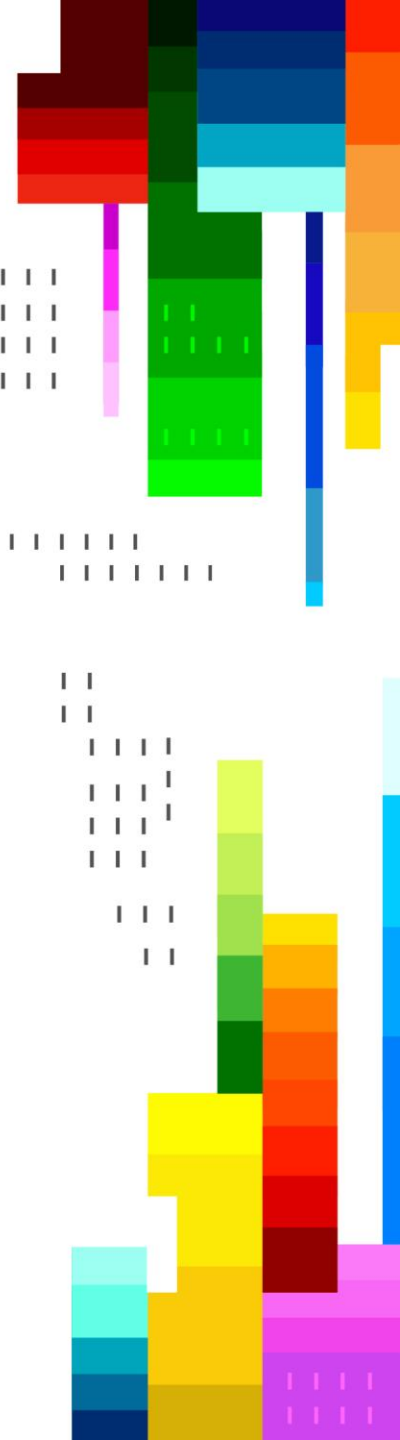




MICROSOFT BUILD



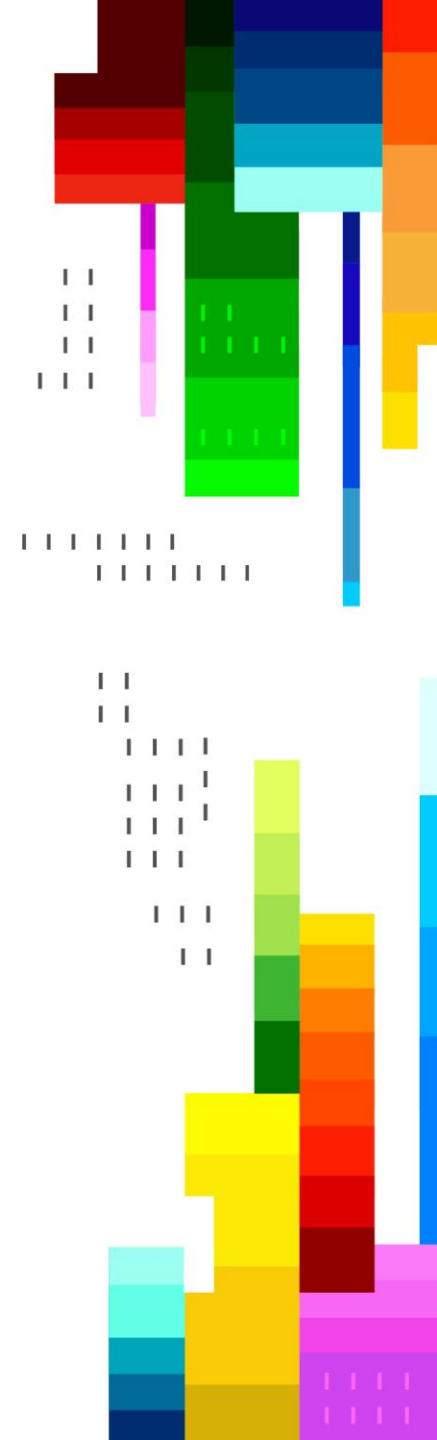


EMU STATION 1 · WORKING WITH MODELS
MODEL SELECTION AND ROUTING

Model Router

in Foundry Models

One endpoint. The best fit model, every time.



What is the Model Router?



A purpose-built, trained model that reads **every prompt in real time** and routes it to the **best-fit LLM**.



Auto-navigates the Foundry model catalog

Picks the best model based on task complexity and your cost/quality preferences.



Routing modes & model subsets

Balanced, Cost, or Quality. Restrict the pool to specific models for compliance or budget.



Foundry Agent Service support

Same routing benefits in agentic, multi-step tool-calling workflows.



Enterprise reliability of Azure

Built-in security, automatic failover, observability, and data-zone boundaries.

A four-stage routing pipeline



A compact ML classifier — not an LLM — predicting which model performs best for each prompt at millisecond overhead.

01



Prompt analysis

Prompt complexity, reasoning depth, Task type, tool/ functional calling needs, and context length and structure—into an internal representation of intent and difficulty.

02



Quality estimation

Prompts that any capable model can handle well vs. prompts that demand deeper reasoning, nuanced judgment. The difficulty-aware routing allows model router to save costs without sacrificing quality.

03



Purposeful selection

Balanced mode is default. In Cost mode, most traffic routes to smaller, cheaper escalating to larger models only when needed. In Quality mode, frontier and high-capability models handle most traffic.

04



Dispatch

Forwards the prompt to the selected model. End-to-end routing overhead: milliseconds — negligible vs. LLM inference. Prompts are never stored.

Automatic model selection in real time



01

Prompts arrive

Requests of varying complexity hit a single router endpoint.

02

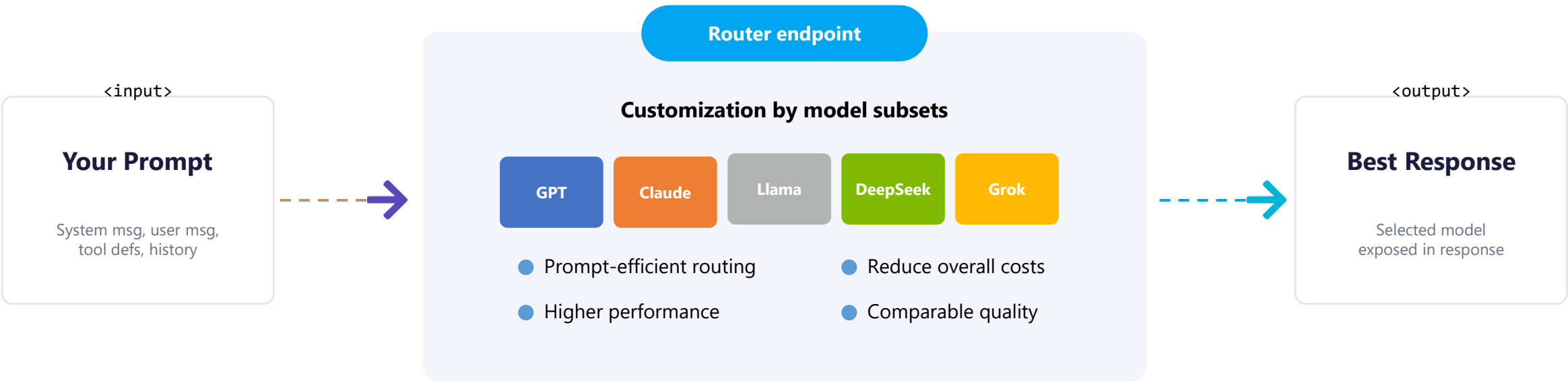
Router analyzes

Intelligent classification maps the prompt to the best-fit model.

03

Response returns

Transparent, cost-effective output — chosen model is always disclosed.



28 models, 5 providers

OpenAI 18 models

gpt-5
gpt-5-mini
gpt-5-nano
gpt-5-chat
gpt-5.2
gpt-5.2-chat
gpt-5.3-chat
gpt-5.4-nano
gpt-5.4-mini
gpt-5.4
gpt-5.5
gpt-4.1
gpt-4.1-mini
gpt-4.1-nano
gpt-4o
gpt-4o-mini
o4-mini
gpt-oss-120b

Anthropic 5 models

claude-haiku-4.5
claude-sonnet-4.5
claude-opus-4.1
claude-opus-4.6
claude-opus-4.7

Meta 1 models

Llama-4-Maverick-17B-128E-Instruct-FP8

DeepSeek 2 models

Deepseek-V3.1
Deepseek-V3.2

xAI 2 models

grok-4
grok-4-1-fast-reasoning

Tune the cost-quality tradeoff



Three modes adjust the quality band — the tolerance for routing to a cheaper model. Even Quality mode doesn't blindly send everything to the most expensive option.



Cost

Wide quality band (~5-6%)

Aggressively favors cheaper models. Nano-class handles the vast majority of traffic.



Balanced

Narrow band (~1-2%)

Picks the most cost-effective eligible model. Spreads traffic across tiers by difficulty.

DEFAULT



Quality

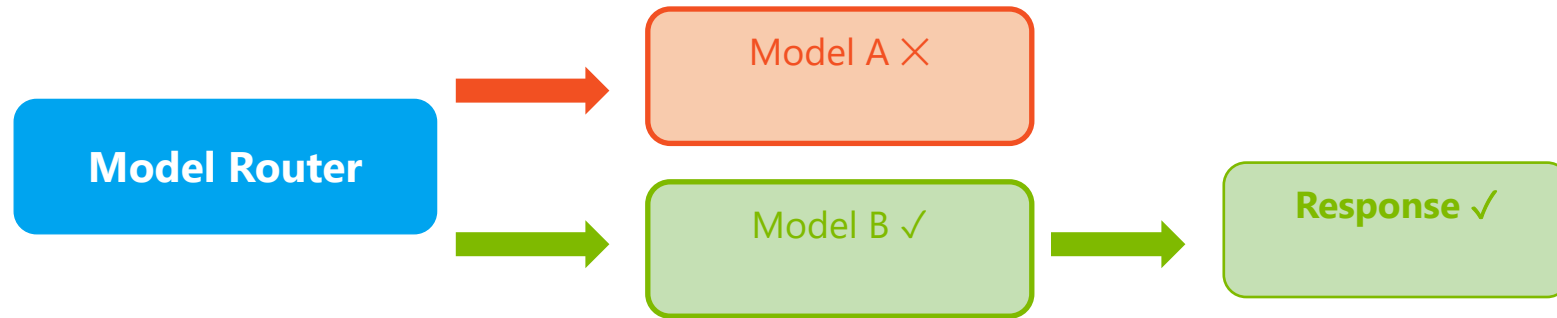
Highest-scoring model

Ignores cost. Frontier models for hard prompts; cheaper still chosen when equivalent.

Up to **76%** cost savings when compared to the most expensive model.
Based in internal data

Automatic failover

Built-in resilience — transient issues never disrupt your application



Why it matters

This resolves the tension: customers get the full breadth of the model pool with platform-grade reliability. The router becomes self-healing — endpoint instability is absorbed transparently.

How it works

- ✓ Zero configuration — failover enabled by default
- ✓ Transparent to your app — response arrives normally
- ✓ Routing mode still applies during failover
- ✓ Model subset = your fallback pool (select 2+ models)

Policy enforcement across the routing stack

Governance applies at the router and cascades to every model in your selected subset

01

Router Governance

Azure Policy definitions apply to Model Router as a single deployment point. One policy controls access, regions, and compliance for the entire routing endpoint.

Applies across portal, CLI, SDK, and ARM/Bicep — a single rule governs the entire routing surface.

02

Subset Model Compliance

Every model in your configured subset inherits and enforces the same policy constraints set at the router level.

Content filtering, data residency, and access controls are honored at the individual model level — not just at the routing layer.

03

Enterprise Impact

60%+ cost and latency savings with full compliance — no tradeoff between performance and governance.

One deployment surface for all 28 models across 6 providers — consistent audit trail and compliance posture.

The screenshot shows the 'Deploy model-router' interface. It includes a 'Deployment information' section with a 'Deployment name' field set to 'model-router' and a 'Deployment type' dropdown set to 'Global standard'. Below this is a warning banner: 'Models blocked by IT Admin 14 models have been blocked for use.' The 'Model router configurations' section is expanded, showing 'Routing mode' set to 'Balanced', 'Model router version' set to '2025-05-08 (Default)', and 'Models subset' options. The 'Models subset' section has two radio buttons: 'Route to all supported models' (unselected) and 'Route to a subset of models' (selected). Below these is a toggle for 'Show available models only'. A table lists models and their versions, with some marked as 'Blocked by IT admin'.

Model	Model version
☐ gpt-5 Blocked by IT admin	2025-08-07
☐ DeepSeek-v3.2 Blocked by IT admin	2025-08-07
☐ gpt-5-mini	2025-08-07
☐ gpt-5-nano Blocked by IT admin	2025-08-07
☐ gpt-5-chat Blocked by IT admin	2025-08-07
☐ gpt-4.1	2025-04-14
☐ gpt-4.1-mini	2025-04-14
☐ gpt-4.1-nano	2025-04-14
☐ o4-mini	2025-04-14

POLICIES CASCADE FROM ROUTER TO EVERY MODEL IN YOUR SUBSET

Fine-tuning: Bring your own training data



~10%

improvement in routing quality
with improved cost efficiency over
default router (early tests)

Why it matters

Every workload has different **quality, cost, and latency tradeoffs**. The default router can't account for your specific domain.

Customers already see **60%+ cost and latency savings** from default routing — customization pushes this further on production traffic.

Self-service workflow

Familiar fine-tuning SDK — no new tools, no data-science expertise required.



Select base model

01

Choose "model router" as the base for fine-tuning.



Upload training data

02

Provide your own labeled prompts and routing preferences.



Submit job

03

Hyperparameters auto-configured — kick off and walk away.



Deploy via Global Standard

04

Fine-tuned router becomes your endpoint, tailored to your domain.

Benchmark the router on your prompts

●● Model-Router-Auto-Evaluation:

An open-source Python toolkit that closes the gap left by built-in Foundry benchmarks — evaluate model router against any baseline on **quality, cost, and latency**.

One command, three outputs



HTML dashboard

Self-contained, interactive — open in any browser.



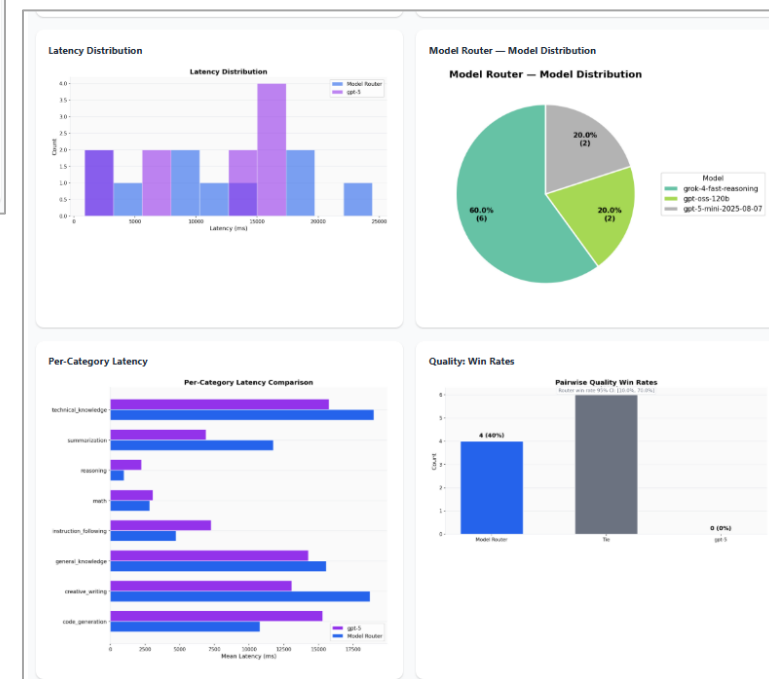
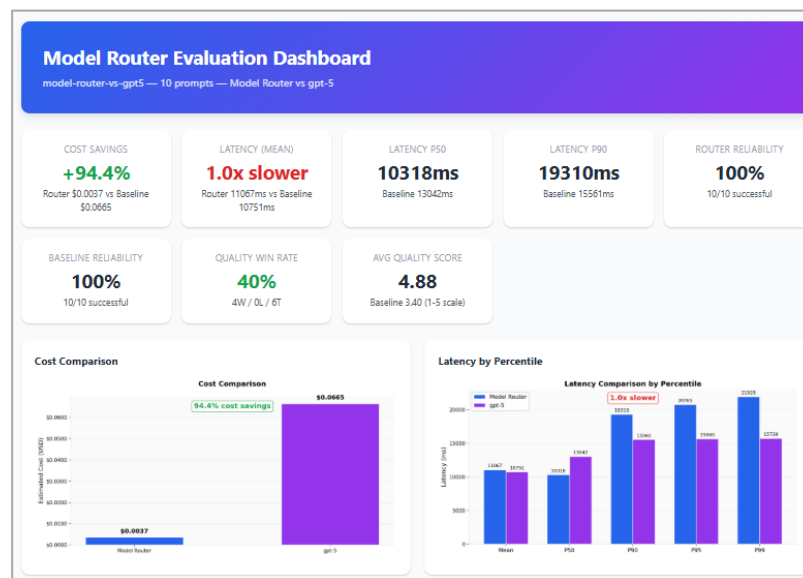
Markdown report

Shareable summary for PRs, docs, decision logs.



Machine-readable results

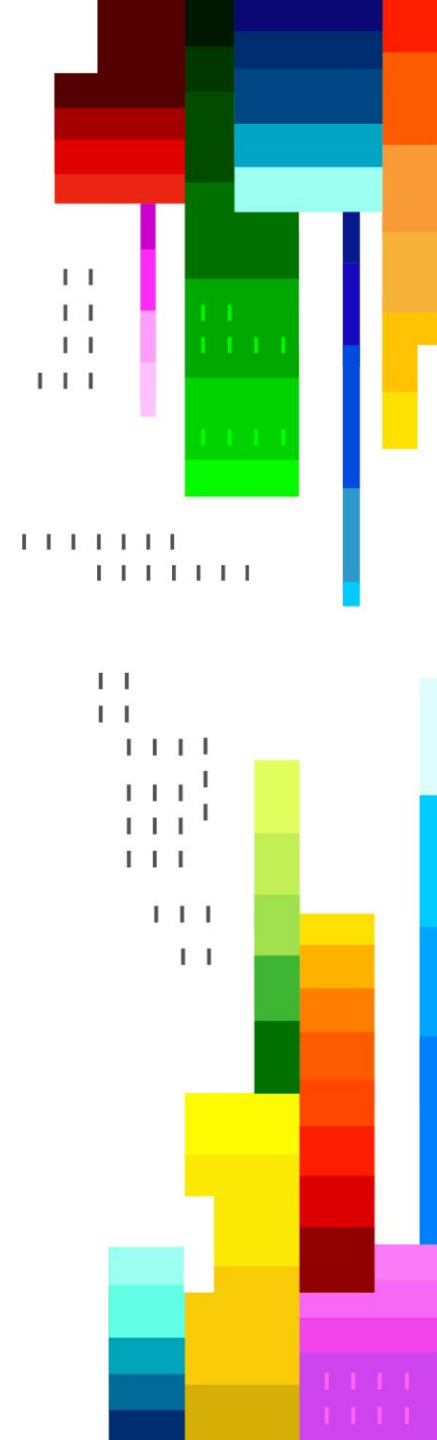
JSON output for CI/CD pipelines and custom analysis.





Agents + Model Router Lab Results

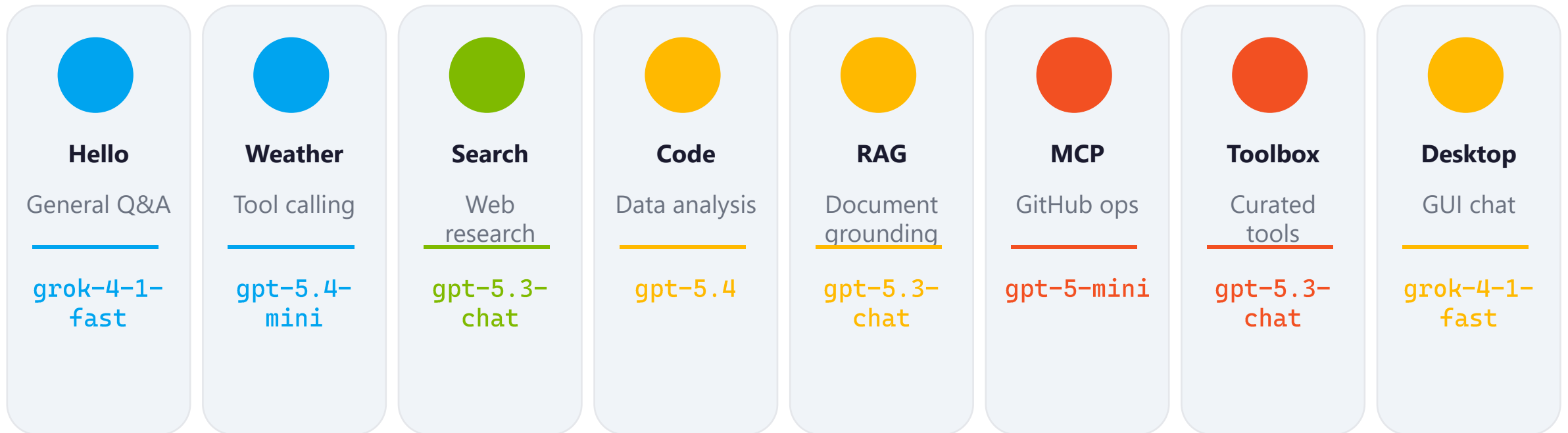
<https://aka.ms/modelrouter-agentslab>



Multiple agents, one model deployment

●● No model pinning, no routing hints — the only variables are tools, system prompt, and query.

Model Router



Five routing patterns observed

●● Consistent behavior across 28 queries reveals the router's decision logic.

01

Fast models for simple facts

Simple factual recall routes to grok-4-1-fast. Cost savings are immediate.



02

Mini models for tool orchestration

Mechanistic tool calls use gpt-5-mini or gpt-5.4-mini — a fraction of frontier cost.



03

Full models for complex reasoning

Research, RAG synthesis, and multi-step operations route to gpt-5.3-chat.



04

Specialized models for specific tasks

Summarization always routes to gpt-5.2-chat, regardless of the agent.



05

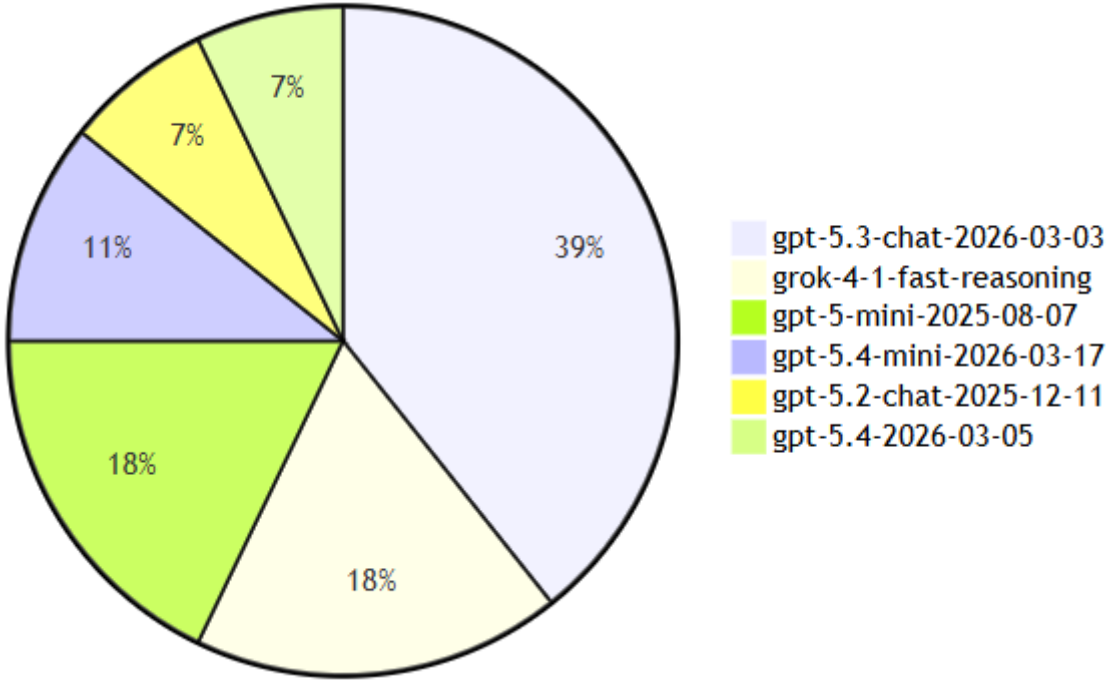
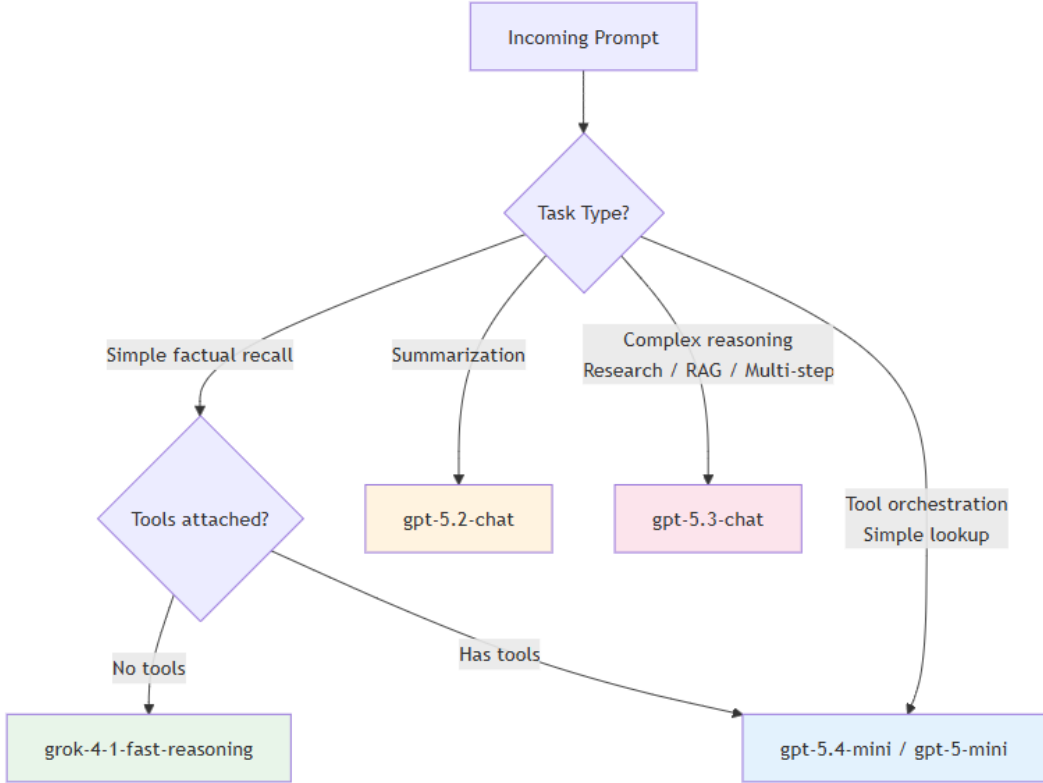
Multi-model conversations

Different turns in the same conversation can use different models. Per-request, not per-session.



How the router decided

●● Observed routing behavior across all queries — the emergent decision tree.



What this means for your agents

●● *Model Router transforms multi-agent architectures — zero config, maximum intelligence.*

01



Zero model pinning

Set `MODEL_DEPLOYMENT=model-router` and let the system optimize. No per-agent configuration needed.

02



Cost without compromise

61% of queries use cheaper models with no quality loss. Frontier capacity reserved for complex work.

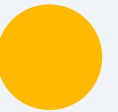
03



Automatic intelligence

Each prompt gets the optimal model. No escalation rules, no manual routing, no prompt analysis code.

04



Auto updates

As new models join the pool, the router adapts. Your agents benefit from improvements with zero changes.

One endpoint · Multiple agents · Zero routing code · Per-prompt intelligence · Immediate cost savings

When to choose Model Router



USE MODEL ROUTER WHEN ...



Diverse workloads

Mix of simple and complex prompts benefits most from intelligent routing.



Cost optimization matters

Small models handle easy prompts; frontier models reserved for hard ones.



Latency-sensitive UX

Chatbots, customer support — lower average latency across the traffic mix.



Building agents

Fast models for classification subtasks, reasoning models for analysis.

USE DIRECT DEPLOYMENT WHEN ...

Same model on every request · Model-specific parameters · Fine-tuned domain models outside the router's pool.

“

We saw a dramatic and immediate reduction in end-user latency, from a volatile 10–20 seconds down to a stable 3–5 seconds. This wasn't just a marginal improvement; it was an (up to) 85% reduction in average response time, which directly translates to a better user experience.

”



Gustavo Azambuja
Senior AI Architect

Deploy from Foundry today

Start with Balanced mode

Observe the routing distribution in Azure Monitor for two weeks before tuning.

Use model subset as a compliance gate

Encode security-team approval; new models won't appear without explicit opt-in.

Monitor routing distribution

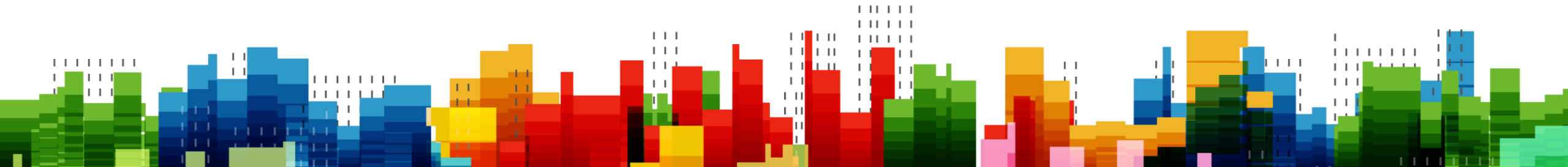
Use model subset as a compliance gate

Pick ≥ 2 models for failover

Single-model subsets defeat the purpose. Handle rate limits client-side with backoff.

Deploy from Microsoft Foundry

<https://aka.ms/trymodelrouter>



Microsoft Foundry Session Guide



Working with Models

BRK230
Build smarter AI systems in Foundry
as models and costs evolve

BRK231
Deploy. Observe. Learn. Reinforcement
learning for production agents

BRK232
Train and deploy custom OSS
reasoning models with Foundry

Responsible AI

BRK250
Govern open-source AI agents,
any framework, any scale

BRK252
From observability to ROI for
AI agents on any framework

BRK251
Build secure and enterprise-ready
agents with Agent 365

Apps & Agents

BRK241
From prototype to production:
build and run agents at scale

BRK240
Build context-aware agents:
from data to decisions

BRK246
Foundry IQ: Fuel agents with enterprise
knowledge and agentic retrieval

BRK243
Claw and agent harness
in Microsoft Foundry

BRK242
Turn your agents into action:
connect tools, APIs, and documents

Agents League Hackathon



Creative
Apps



Reasoning
Agents



Enterprise
Agents

Learn more



May 19th – June 14th

Register now, submit your
project and compete for prizes!

aka.ms/agentsleague/aisf